Seminar: Enterprise AI

Seminar Paper

Classical NLP for Sentiment and Aspect-Based Analysis of Amazon Reviews

A Study of the Trade-off Between Predictive Performance and Interpretability

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Ivelina Yaneva

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Julius-Maximilians-Universität Würzburg
Chair of Information Systems and Business Analytics
Supervisor: Viet [Nachname]

Online reviews pair a star rating with free text that explains it, offering a continuous, survey-free feedback signal — if the text can be read at scale. This paper builds an end-to-end classical NLP pipeline (no large language models) that turns 87,713 Amazon reviews from two subscription categories into structured opinion data, and studies the central trade-off between predictive performance and interpretability. For document-level polarity (RQ1) we compare a majority-class baseline against Naive Bayes, Logistic Regression and gradient boosting on TF-IDF features, under binary and ternary framings and with explicit class-imbalance handling. Logistic Regression with balanced class weights is the strongest model (binary macro-F1 0.87) while remaining fully interpretable; the headline accuracy of a trivial baseline (69%) is shown to be misleading because it never identifies a single negative review. A ternary framing exposes the hard ceiling of the task: the minority 3-star (neutral) class collapses to 0.4% recall without rebalancing. For aspect-based sentiment (RQ2) we segment reviews into sentences, detect seven aspects with a keyword lexicon and a noun-phrase view, and score each aspect-mentioning sentence with VADER — a lexicon scorer chosen because the document classifier is out-of-distribution on single sentences. Across both categories, *billing* (cancel/refund/charged) is the clear pain point, while *content* and *price* are the most positive aspects. Crucially, the strongest negative coefficient of the interpretable classifier is the token *cancel* — the same billing signal RQ2 surfaces independently, a convergence of two methods on one business conclusion.

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1. Introduction

Every day, millions of customers leave reviews on platforms such as Amazon, Yelp and Google Maps. Each review couples a structured signal — an overall star rating — with a block of free text explaining the reasons behind it. For a company, this is a continuous stream of customer feedback that requires no surveys; the difficulty is reading it at scale. *Sentiment analysis*, the task of inferring opinion polarity from text (Pang and Lee, 2008), makes this tractable, and *aspect-based sentiment analysis* (ABSA) refines it by attributing sentiment to specific facets such as delivery, price or quality (Pontiki et al., 2014).

This paper deliberately restricts itself to *classical* natural language processing and machine learning — bag-of-words features and linear or tree-based classifiers — without any large language model. This is not only a constraint but a research lens: classical models are cheap, reproducible and, above all, *interpretable*, and the central question of this work is how much predictive performance, if any, that interpretability costs. We study two Amazon review categories from the 2023 release (Hou et al., 2024) and pursue one overarching and two specific research questions.

Main RQ
What is the trade-off between predictive performance and interpretability when classical machine learning is applied to review sentiment and product-aspect extraction?

RQ1
How accurately do classical models classify review-level sentiment, how does class imbalance affect their evaluation, and which review types are hardest to classify?

RQ2
Which product aspects do customers discuss, and how do aspect sentiment profiles differ between two product categories?

Contributions. (i) A reproducible classical pipeline (cleaning, TF-IDF features, three classifier families, evaluation) with an honest treatment of class imbalance, showing that accuracy is the wrong headline metric and that a balanced Logistic Regression matches gradient boosting while staying interpretable. (ii) An unsupervised, sentence-level ABSA pipeline using a domain-adapted aspect lexicon and VADER, with a robustness check against the document classifier. (iii) A cross-method finding: the most negative linear-model coefficient and the most negative VADER aspect independently identify *billing* as the dominant source of dissatisfaction for subscription products.

2. Background and Related Work

2.1. Sentiment analysis

Opinion mining and sentiment analysis are surveyed comprehensively by Pang and Lee (2008), who frame polarity classification as a supervised learning problem over textual features and discuss the role of domain, negation and subjectivity. A long line of work treats document-level polarity as text classification with bag-of-words or n-gram features (Maas et al., 2011; Manning et al., 2008); we follow this classical recipe rather than dense neural representations.

2.2. Aspect-based sentiment analysis

Pontiki et al. (2014) formalise ABSA in SemEval-2014 Task 4, distinguishing aspect-term extraction, aspect-category detection and per-aspect polarity, and provide gold-annotated benchmarks. Their setting is supervised; in the absence of aspect-level annotations for the Amazon categories studied here, we approximate the same goals with an unsupervised lexicon-plus-VADER approach (Section 4.7), which trades annotation cost for transparency.

2.3. Features, models and lexicon scoring

TF-IDF weighting over unigrams and bigrams is the standard sparse text representation (Manning et al., 2008). We compare three classifier families that span the bias–variance and interpretability spectrum: Multinomial Naive Bayes (a generative linear baseline), Logistic Regression (a discriminative linear model whose weights are directly readable), and gradient-boosted decision trees via LightGBM (Ke et al., 2017). For sentence-level sentiment we use VADER (Hutto and Gilbert, 2014), a rule-based valence lexicon designed for short, informal text that needs no training data. All models are implemented with scikit-learn (Pedregosa et al., 2011), spaCy (Honnibal and Montani, 2017) and NLTK (Bird et al., 2009).

2.4. Why classical, and the neural contrast

Transformer sentence encoders such as Sentence-BERT (Reimers and Gurevych, 2019) achieve strong results on semantic tasks but are comparatively opaque and resource-intensive. For an operational feedback dashboard, the value of a model that can answer "*which words drove this prediction?*" is high. Our study therefore quantifies what is given up by staying classical, and finds (Section 7) that on these high-dimensional sparse features the interpretable linear model is not outperformed by the more complex tree ensemble.

3. Dataset and Exploratory Analysis

3.1. Dataset and category choice

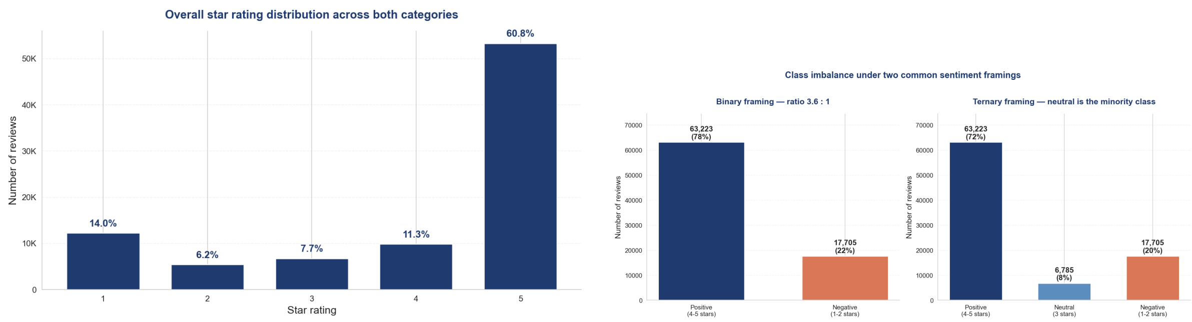
We use the Amazon Reviews 2023 dataset (Hou et al., 2024). The task asks for the smallest category; because RQ2 requires a *cross-category* comparison, we deliberately analyse *two* categories: **Subscription Boxes** (16,216 reviews, 641 distinct items — among the smallest categories) and **Magazine Subscriptions** (71,497 reviews), for a combined **87,713** reviews. Each review provides a 1–5 star `rating`, a `title`, free-text `body`, a `timestamp`, a `verified_purchase` flag and a `helpful_vote` count. Table 1 summarises the key statistics.

Table 1: Key dataset statistics from the exploratory analysis.

Statistic	Value
Total reviews	87,713
Subscription Boxes	16,216
Magazine Subscriptions	71,497
Time span	2001–2023
Verified-purchase share	82.7 %
Mean / median review length	40 / 22 words
5-star share	60.8 %
1-star share	14.0 %
Binary imbalance (pos:neg)	3.57 : 1
Neutral (3-star) share	7.7 %
Vocabulary size (min_df = 5)	6,308

3.2. Star distribution and class imbalance

The rating distribution is strongly positive-skewed: 60.8% of reviews are 5-star and only 14.0% are 1-star (Figure 1). Mapping ratings to sentiment (Section 4.1) yields a 3.57 : 1 positive-to-negative imbalance in the binary framing and a small 7.7% neutral class in the ternary framing (Figure 2). This imbalance is the single most important property for evaluation: a classifier that always predicts "positive" already attains high accuracy while being useless, which is why we report macro-F1 and per-class recall throughout.



(a) Overall star distribution.

(b) Sentiment-class imbalance.

Figure 1: Star ratings are heavily skewed toward 5 stars, producing a substantial positive/negative imbalance and a small neutral class.

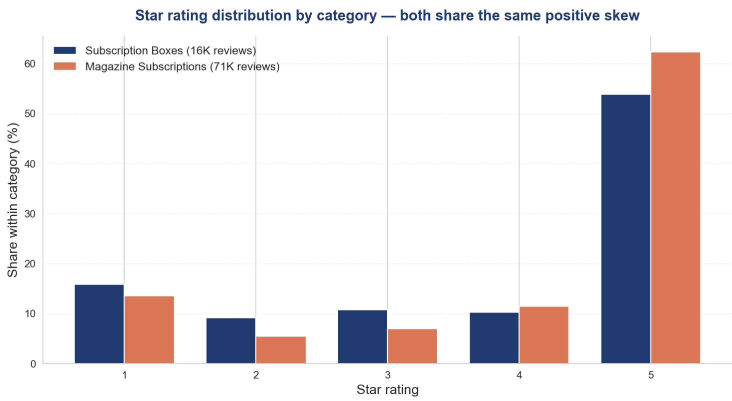
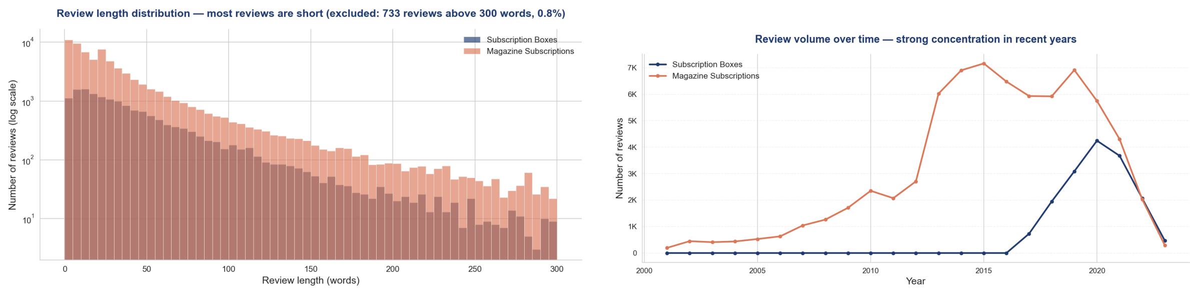


Figure 2: Per-category star distribution: both categories share the positive skew, confirming the imbalance is not an artefact of one category.

3.3. Review length and vocabulary

Reviews are short: a median of 22 words and a mean of 40 (Figure 3). Short texts carry few features, which both favours sparse linear models and motivates the sentence-level treatment in RQ2. A whitespace-tokenised vocabulary on a 20K-review sample, filtered at min_df = 5, contains roughly 6,300 terms.



(a) Review-length distribution (words).

(b) Review volume over time.

Figure 3: Most reviews are short, and review volume is concentrated in recent years — which motivates a chronological rather than random train/test split.

3.4. Discriminative terms

A weighted log-odds analysis with an informative Dirichlet prior (Monroe et al., 2008) identifies the terms most associated with each sentiment class. Rendered as word clouds (Figure 4), the positive class is dominated by generic affect (*love, great, enjoy, wonderful*), while the negative class is strikingly concrete: *cancel, refund, subscription, ad, advertisement, charge, renewal*. Already at the descriptive stage, the data points to billing and advertising as the negative themes that the modelling later confirms.

Figure 4: Most discriminative terms per sentiment class (size ∝ weighted log-odds z-score). The negative cloud is a billing/advertising cluster.

4. Methodology

4.1. Sentiment labelling

Star ratings are the natural supervision signal but exist only per review, so RQ1 operates at the document level. We use two framings. The **binary** framing maps 1–2 stars to *negative*, 4–5 to *positive* and drops 3-star reviews; the **ternary** framing keeps 3 stars as *neutral*. Binary leads the analysis; ternary is reported to probe the hard, mixed-sentiment middle.

4.2. Text preprocessing

The body text is lowercased and stripped of HTML, URLs and non-alphabetic characters, then tokenised, stopword-filtered and lemmatised with spaCy (`en_core_web_sm`). A small number of reviews (275) reduce to an empty string and are dropped before modelling, as they carry no bag-of-words signal. The cleaned text feeds only the TF-IDF classifiers of RQ1; VADER in RQ2 deliberately consumes the *raw* text, because its valence rules exploit capitalisation, punctuation and intensifiers.

4.3. Chronological train/test split

Because review volume is concentrated in recent years, a random split would let a model train on future reviews. We therefore sort by timestamp and use the oldest 80% for training and the newest 20% for testing, performed *within each category* so that both categories appear on both sides — preserving the cross-category comparison RQ2 needs while making evaluation mimic deployment.

4.4. Features

We build TF-IDF vectors with sublinear term-frequency scaling, min_df = 5 and a vocabulary cap of 20,000. Two configurations are compared: unigrams, and unigrams+bigrams; the vectoriser is fit on the training split only.

4.5. Models and imbalance handling

We evaluate a **majority-class baseline** (always predicts positive), **Multinomial Naive Bayes**, **Logistic Regression** and **LightGBM** gradient boosting (Ke et al., 2017). Imbalance is handled two ways: Logistic Regression and LightGBM use `class_weight=balanced`; Naive Bayes, which has no class-weight parameter, is trained on a randomly undersampled balanced set. Each model is run with and without handling so the effect is visible.

4.6. Evaluation

We report accuracy (for reference), macro-averaged F1, weighted F1 and per-class recall, plus confusion matrices. Macro-F1 and minority-class recall are the primary metrics under imbalance. Error analysis ranks the most confidently wrong predictions to characterise the hardest reviews.

4.7. Aspect-based pipeline (RQ2)

A review-level label cannot express that one review praises content but criticises price, so RQ2 works below the review. Each review is split into sentences; aspects are detected per sentence using a curated keyword lexicon of seven aspects — *delivery, packaging, quality, price, content, customer service* and *billing* — the last two added because the EDA surfaced content and billing/renewal vocabulary prominently. A complementary, lexicon-free noun-phrase view (spaCy noun chunks) provides data-driven aspect candidates. Each aspect-mentioning sentence is scored with VADER (Hutto and Gilbert, 2014), and scores are aggregated per review, per aspect and per category.

Why VADER rather than the RQ1 classifier. The RQ1 classifier is trained on whole reviews; applying it to single sentences is an out-of-distribution use, since sentences are far shorter with a different feature distribution. VADER needs no training and is built for short text, so it is the *primary* sentence-level scorer. The document classifier is retained only as a *robustness check*: we measure how often the two methods agree rather than assuming either is ground truth (Section 6.4).

5. Results: Document-Level Sentiment (RQ1)

5.1. Binary framing

Table 2 reports the binary results on the held-out test set. The majority baseline reaches 68.9% accuracy but macro-F1 0.408 and **zero** recall on the negative class — a vivid demonstration that accuracy hides total minority failure. Without imbalance handling, the three models inherit a milder version of this skew (Naive Bayes recovers only 55% of negatives). Switching on balanced weighting or undersampling lifts negative recall to 0.87–0.89. **Logistic Regression with balanced weights is the best overall model** (accuracy 0.885, macro-F1 0.870, both classes recalled near 0.88); here, balancing even *improves* accuracy because the unbalanced model was sacrificing too many negatives. Figures 5 and 6 visualise the recovery.

Table 2: RQ1 binary model comparison (test set). Macro-F1 and per-class recall are primary; accuracy is reference only. Best macro-F1 in **bold**.

Model	Accuracy	Macro-F1	$\text{Recall}_{\text{pos}}$	$\text{Recall}_{\text{neg}}$
Majority baseline	0.689	0.408	1.000	0.000
Naive Bayes	0.840	0.787	0.971	0.550
Logistic Regression	0.871	0.838	0.961	0.672
Gradient Boosting	0.857	0.817	0.961	0.628
Naive Bayes (resampled)	0.860	0.845	0.844	0.894
Logistic Reg. (balanced)	**0.885**	**0.870**	0.889	0.875
Gradient Boosting (balanced)	0.850	0.834	0.843	0.867

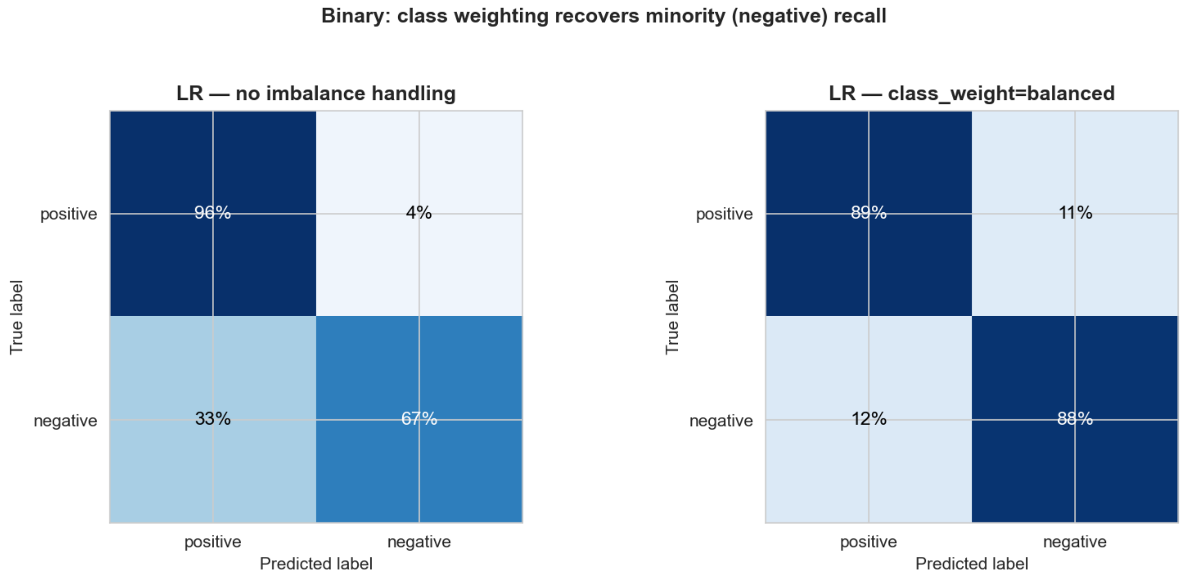


Figure 5: Binary confusion matrices (row-normalised). Balanced class weighting turns a model that misses a third of negatives into one that recalls them evenly.

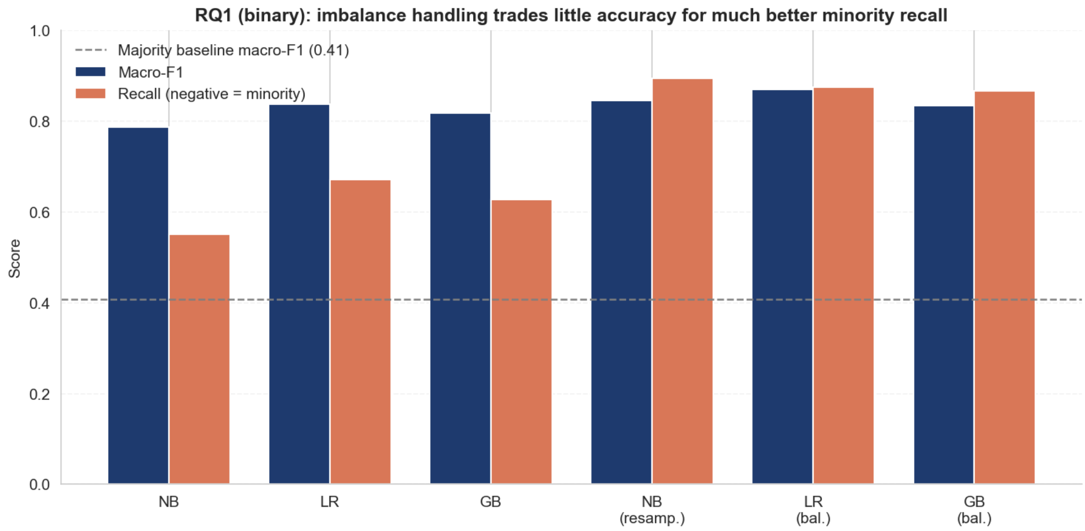


Figure 6: Macro-F1 and minority-class (negative) recall, with and without imbalance handling. The dashed line is the majority-baseline macro-F1.

5.2. Ternary framing

Adding the 3-star neutral class makes the task markedly harder (Table 3). Without handling, Naive Bayes predicts the neutral class with 0.4% recall — a textbook minority collapse — and even Logistic Regression reaches only 9%. Balancing raises Logistic Regression's neutral recall to 42% and Naive Bayes' (resampled) to 50%, but at a genuine accuracy cost (0.81 $\rightarrow$ 0.77 for LR). This is the central RQ1 trade-off: the 3-star class never becomes easy because such reviews genuinely mix positive and negative content, which directly motivates the sentence-level view of RQ2.

Table 3: RQ1 ternary model comparison (test set). The neutral (3-star) recall column shows the minority collapse and its partial recovery.

Model	Acc.	Macro-F1	Rec_{pos}	Rec_{neu}	Rec_{neg}
Majority baseline	0.639	0.260	1.000	0.000	0.000
Naive Bayes	0.780	0.508	0.970	0.004	0.551
Logistic Regression	0.813	0.587	0.954	0.087	0.681
Gradient Boosting	0.799	0.571	0.957	0.083	0.626
Naive Bayes (resampled)	0.714	0.596	0.742	0.499	0.703
Logistic Reg. (balanced)	0.767	**0.625**	0.832	0.423	0.708
Gradient Boosting (balanced)	0.733	0.603	0.782	0.449	0.695

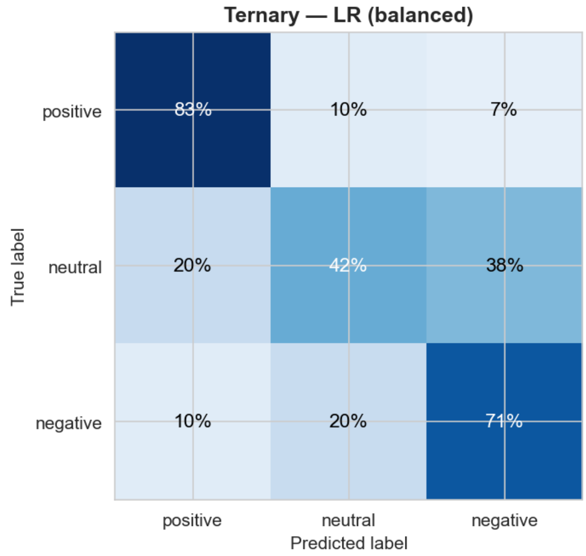


Figure 7: Ternary confusion matrix (Logistic Regression, balanced). Neutral reviews are most often confused with the adjacent positive class.

5.3. Feature representation

Adding bigrams roughly doubles the vocabulary (8,838 → 20,000 features) for a small but real gain in macro-F1 (0.861 → 0.870; Table 4). Bigrams capture negations and short phrases ("not worth", "highly recommend") that unigrams miss. For a strictly leaner, interpretability-first setup, unigrams remain defensible.

Table 4: Feature comparison (Logistic Regression, balanced, binary).

Features	# Features	Accuracy	Macro-F1
TF-IDF unigram	8,838	0.876	0.861
TF-IDF uni+bigram	20,000	0.885	0.870

5.4. Interpretability

A linear TF-IDF model exposes a signed weight per feature. Table 5 and Figure 8 list the strongest. The positive weights are generic affect; the negative weights are dominated by *disappointed, cancel, disappointing, ad, advertisement* and the bigram *cancel subscription*. That the single strongest negative driver is a *billing* word, not a product-quality word, is the key qualitative result of RQ1 — and it directly anticipates RQ2.

Table 5: Strongest Logistic-Regression coefficients (binary, balanced).

Positive term	Weight	Negative term	Weight
love	+14.6	disappointed	−7.9
great	+9.0	cancel	−7.7
enjoy	+8.0	disappointing	−6.3
excellent	+6.4	ad	−5.6
perfect	+5.7	poor	−5.5
amazing	+5.5	boring	−5.2
easy	+5.4	waste	−5.0
favorite	+5.3	advertisement	−4.8

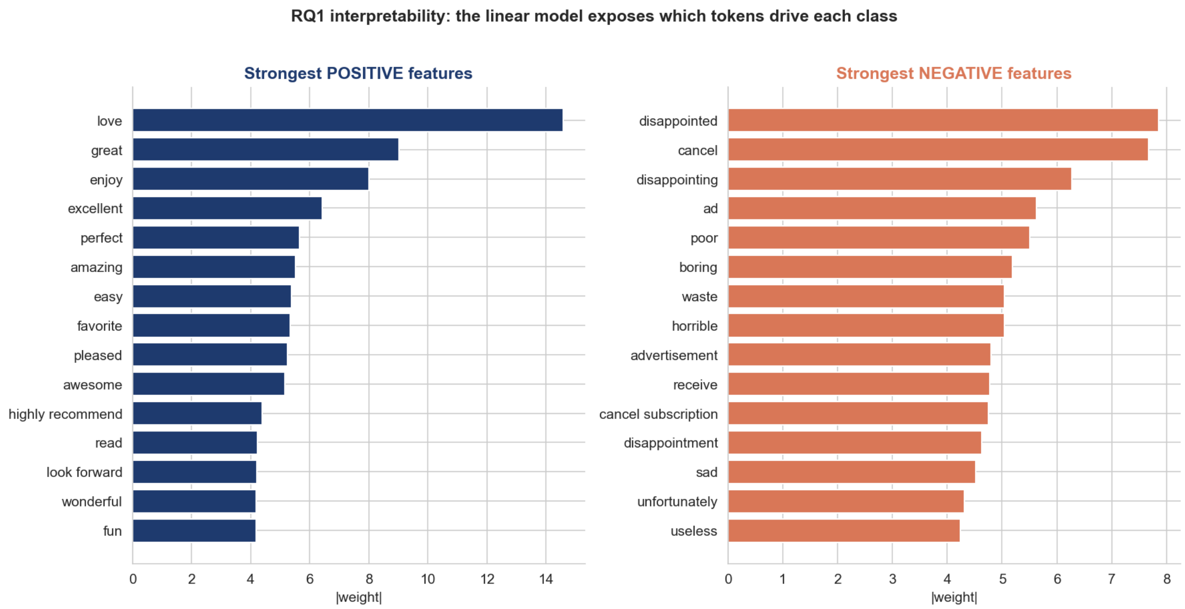


Figure 8: The linear model is fully auditable: each token carries an explicit, signed contribution to the positive/negative decision.

5.5. Error analysis: the hardest reviews

Ranking the most confidently wrong predictions reveals that many are not model failures but *label noise*: reviews whose star rating disagrees with their text. Examples include 5-star reviews reading "*Very disappointed ... Cancelled*", "*ad after ad ... overpriced*" and "*Waste of money*", and a 1-star review reading simply "*Perfect*". The model's text-based judgement is arguably correct in these cases; the star rating is the unreliable signal. This both bounds achievable accuracy and reinforces the value of reading the text rather than trusting the star alone.

6. Results: Aspect-Based Sentiment (RQ2)

6.1. What customers discuss

Aspect mention rates differ sharply by category (Figure 9, Table 6). *Content* dominates Magazine reviews (63.5% mention it), whereas *packaging* dominates Subscription-Box reviews (46.5%): the physical unboxing matters for boxes, the editorial substance for magazines.

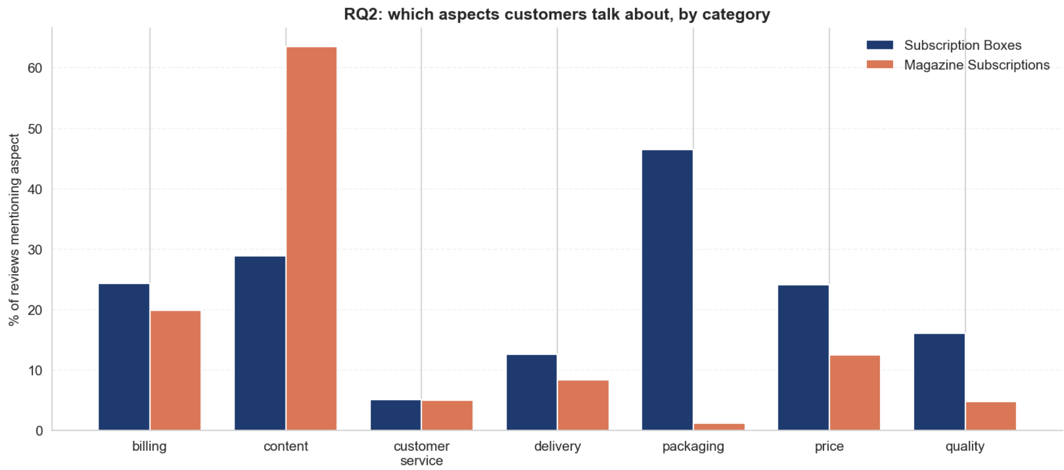


Figure 9: Share of reviews mentioning each aspect, by category. *Content* drives magazines; *packaging* drives subscription boxes.

6.2. Aspect sentiment

The radar and bar views (Figures 10, 11) and Table 6 agree on one striking pattern: in *both* categories, *billing* has by far the lowest mean sentiment (compound $\approx$ 0.17–0.18) and the highest negative share (21% for magazines, 28% for boxes), while *content* and *price* are the most positive (compound $\approx$ 0.34–0.38). The sentiment composition per aspect (Figure 11) shows the same: billing carries the largest red (negative) band.

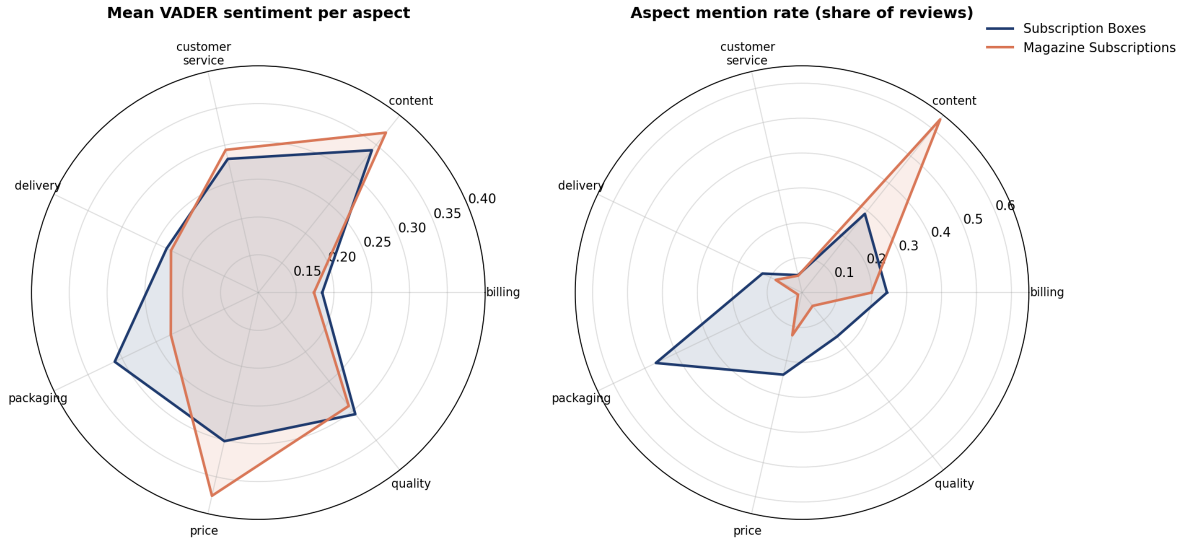
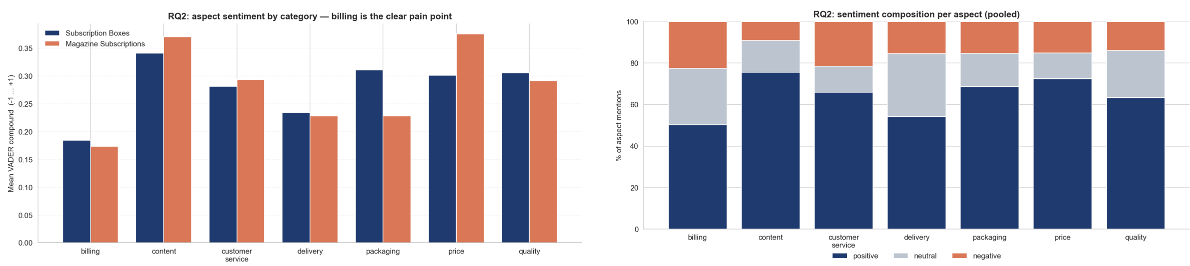


Figure 10: Aspect profiles as radar charts. Left: mean VADER sentiment — both polygons collapse inward at *billing*. Right: mention rate — the categories' different shapes (content vs. packaging).

Table 6: Per-aspect mention rate, mean VADER compound and negative share, by category. Lowest sentiment per category in **bold**.

	Subscription Boxes			Magazine Subscriptions		
Aspect	Mention %	Mean	Neg %	Mention %	Mean	Neg %
billing	24.4	**0.185**	27.8	19.9	**0.174**	21.0
content	28.9	0.341	13.5	63.5	0.371	8.7
customer service	5.1	0.281	20.4	5.0	0.294	21.9
delivery	12.6	0.235	19.4	8.4	0.228	14.2
packaging	46.5	0.311	14.8	1.3	0.229	19.2
price	24.1	0.302	20.1	12.6	0.376	12.9
quality	16.1	0.306	15.0	4.9	0.292	13.1



(a) Mean sentiment per aspect.

(b) Sentiment composition (pooled).

Figure 11: Aspect sentiment as bars and as positive/neutral/negative composition. Billing is the most negative aspect on both views.

6.3. Noun-phrase view

The most frequent noun phrases (Figure 12) independently validate the lexicon: *the box, the toys, the products, the price, the items* for boxes and *this magazine, the articles, subscription* for magazines all map onto the predefined aspects. (One data artefact: a stray `br` token for magazines, a remnant of `<br>` HTML tags in the raw text, which the noun-phrase view uses unmodified; it does not affect the keyword/VADER pipeline.)

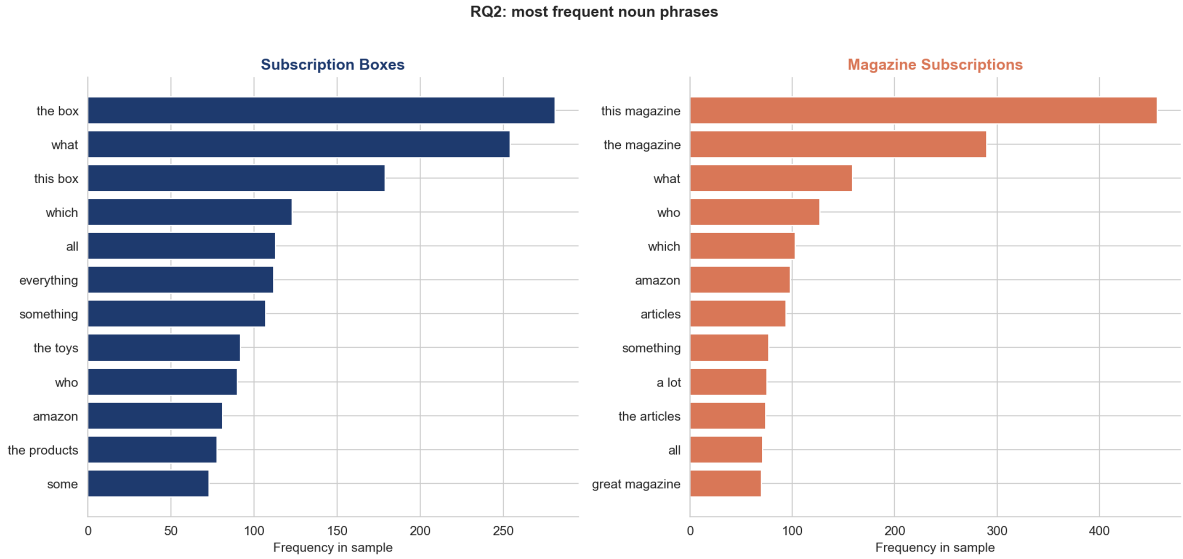


Figure 12: Most frequent noun phrases per category — a data-driven cross-check of the keyword aspect lexicon.

6.4. Robustness: VADER versus the classifier

On a test-set sample, VADER and the RQ1 classifier agree on $\approx 75\%$ of non-neutral aspect mentions (Table 7) — close enough to confirm VADER is not producing noise, divergent enough to matter. Importantly, VADER labels about 22% of mentions *neutral*, a category the binary classifier structurally cannot express. Combined with the out-of-distribution concern, this justifies VADER as the primary sentence-level scorer and the classifier as a sanity check only.

Table 7: Agreement between VADER and the document classifier on non-neutral aspect mentions (test sample).

Aspect	Agreement	n
content	0.803	639
packaging	0.737	114
quality	0.737	76
price	0.730	137
billing	0.690	203
customer service	0.679	56
delivery	0.602	83
Overall	$\approx$**0.75**	1,308

6.5. Qualitative lexicon validation

Because no gold aspect labels exist, we inspected a sample of aspect-assigned sentences (Appendix A). The keyword lexicon assigns aspects correctly in the large majority of cases; the main failure mode is incidental keyword use (e.g. "value" meaning worth-as-virtue rather than price). This, and the single-score-per-sentence limitation, are discussed next.

7. Discussion

Performance versus interpretability (Main RQ). On these high-dimensional, sparse TF-IDF features, the fully interpretable Logistic Regression is the best model under both framings and is *not* outperformed by gradient boosting (binary macro-F1 0.870 vs. 0.834). The intuition is standard: linear models excel when the signal is largely additive over many sparse lexical features, where tree ensembles gain little. The headline finding for the Main RQ is therefore encouraging — for this task, choosing the transparent model costs essentially no accuracy, and it buys a model whose every decision can be traced to specific words (Table 5).

Two methods, one conclusion. The most negative coefficient of the supervised classifier (*cancel*) and the most negative aspect of the unsupervised VADER pipeline (*billing*) identify the *same* business problem from two independent directions. For subscription products, dissatisfaction is concentrated not in product quality but in the billing and cancellation experience — a directly actionable insight.

Label noise and the neutral ceiling. The error analysis showed that star ratings and text sentiment genuinely disagree for a non-trivial set of reviews, and the ternary results showed the 3-star class cannot be reliably separated. Both point to the same truth: the star rating is a noisy, coarse label, and the hardest reviews are the genuinely mixed ones — exactly where aspect-level analysis adds value over a single document label.

Business implications. A practitioner can deploy the balanced Logistic Regression as a transparent polarity monitor, and the lexicon+VADER layer as an aspect dashboard that flags *which* facet drives sentiment. For both subscription categories, the first intervention the data recommends is fixing billing/cancellation friction.

8. Limitations and Future Work

- **Sentence-level attribution** cannot separate two aspects that co-occur in one sentence with mixed sentiment ("great content ... but overpriced"): both inherit the sentence's single VADER score. Clause-level splitting on coordinating conjunctions is the natural refinement.
- **Lexicon and scorer** are hand-built and English-only, and VADER is general-purpose rather than domain-tuned; a domain-adapted lexicon would improve precision.
- **No gold aspect labels** exist, so RQ2 is validated only against a second method and qualitative inspection, not ground truth. A small annotated set (SemEval-style, Pontiki et al., 2014) would enable proper scoring.

- **Star ratings as labels** are noisy, as the error analysis showed, placing a ceiling on supervised accuracy.
- **Category scope:** Subscription Boxes has only 641 distinct items, so its aspect frequencies can be influenced by a few popular boxes.
- **Neural contrast:** a transparent linear model was the goal here; a controlled comparison against a Sentence-BERT classifier (Reimers and Gurevych, 2019) would quantify the interpretability–accuracy frontier more precisely.

9. Conclusion

We built a fully classical, reproducible NLP pipeline that turns 87,713 Amazon subscription reviews into structured opinion data. For document-level polarity, class imbalance — not model choice — is the dominant evaluation concern: a 69%-accurate baseline identifies no negatives, and only after explicit rebalancing do the models recall the minority class well, with an interpretable balanced Logistic Regression the strongest (0.87 macro-F1) and as good as gradient boosting. For aspect-based sentiment, a sentence-level lexicon+VADER pipeline shows that *what* customers discuss is category-specific (content vs. packaging) while *what frustrates* them is shared and concentrated in *billing*. That the interpretable classifier's strongest negative word is *cancel* — the same billing signal — is the study's central result: on this task, interpretability is essentially free, and the transparent model hands the business a directly actionable diagnosis.

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A. Additional material

The repository accompanying this paper contains the full result tables and figures, including per-class precision/recall/F1, the complete top-noun-phrase lists per category, and the sampled aspect-assigned sentences used for the qualitative lexicon validation (`results/tables/`). All notebooks (`01_eda` through `05_paper_figures`) reproduce every figure and table from the raw data.

Declaration of Originality

I declare that I have authored this paper independently, that I have not used other than the declared sources / resources, and that I have explicitly marked all material which has been quoted either literally or by content from the used sources.

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Ivelina Yaneva